Towards a Cartography of Datacentering

A Multimodal Research Agenda for Innovation Studies

Babajide Alamu Owoyele

Research Policy 11th ECR Conference | 27 April 2026

DRIFT, Erasmus University Rotterdam | HPI

The Empirical Moment

Metric	Value
Hyperscaler capex 2026	**$750B**
DC electricity consumption 2025	485 TWh (+17% YoY)
Data center demand growth 2025–26	+50% YoY (Goldman Sachs 2026)
Hyperscaler share of DC capacity 2031	67% of global total

"The largest concentrated infrastructure build of our lifetimes — happening now."

Not a future scenario. Not a policy proposal.

An arena in formation.

Datacentering: the contested, multi-scalar process by which data center infrastructure becomes embedded in energy systems, planning regimes, financial markets, and built environments — reshaping each in turn.

CreditSights (Feb 2026);
IEA Energy & AI Report (Apr 2026);
Synergy Research Group (2025)

Why Now? The Contested Arena

- **Stargate:** $500B / ~7 GW committed (OpenAI, Oracle, SoftBank)
- **PJM electricity:** capacity prices **+833%** (2024/25→2025/26); DC = 40% of auction cost [PJM Dec 2025]
- **Maine:** legislature passes statewide moratorium bill — governor vetoes, Apr 2026 [DCD corpus; Mills veto 24 Apr 2026]
- **Loudoun Co.:** eliminates by-right DC zoning, March 2025 [DCD corpus]
- **DeepSeek:** re-opens Jevons paradox debate on AI compute demand[a]

37,552 DCD articles

2008–2026, headline + intro

1,073 contestation events

Peak contestation: 2025

Trade press = primary event record before academic literature

[a] Jevons paradox: efficiency gains increase, not decrease, total consumption by stimulating demand.

The Gap in Innovation Studies

RP canon covers:

Platforms (Gawer 2014)
GPTs (Brynjolfsson)
Sectoral systems (Malerba)
Transitions (Geels)

Critical DC studies exists:

Big Data & Society, *Convergence*
Edwards et al. 2025; Monstadt 2025
— but outside RP's canon

Missing:

Integrative framework for the *infrastructural + methodological* turn in AI-era innovation studies

No unified approach to:
- mapping contested arenas
- surfacing implicated actors
- operationalizing multimodal evidence

Theory: Arenas of Transformation

After Jørgensen's *arenas of development* (2012, *Res. Policy*)

Flat ontology. Start from *actor constellations* and their collective engagement — not structural tiers.

Negotiation is the fundamental process.

Goal: **characterize what is *emerging*** before the arena stabilises.

+ Clarke (2005): situational maps

Implicated actors: silenced (communities, visa workers) + constructed but absent ("the consumer", "future generations")

The datacentering arena (in formation):

Co-constitutive.
Not hierarchically ordered.

Methodological Framework: Issue Mapping + CITC

Marres (2015) — *ST&HV*

Issue mapping via digital traces.
Trade press, hyperlinks, hashtags as empirical window into issue formation.

Berente et al. (2019); Miranda et al. (2022)

Computationally Intensive Theory Construction

Sampling → synchronic → diachronic → **lexical framing**

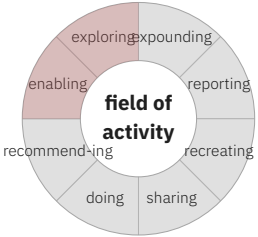
Patterns become contributions *only* once situated in scholarly discourse.

Marres tells us
what to trace

CITC tells us
how to trace it
rigorously enough for RP

Prevents atheoretical pattern-mining.

Registerial Cartography of Datacentering



contestation register

Matthiessen (2015): registerial cartography

Register = functional variety of language according to context of use.

Field of activity = what's going on:

doing	joint social activity
enabling	action modelled via instruction
exploring	public values debated
reporting	events chronicled

Applied to DCD corpus:

Non-contestation: *reporting* + *doing*
Contestation: *enabling* + *exploring*

Verb register shift = governance gap made linguistically visible

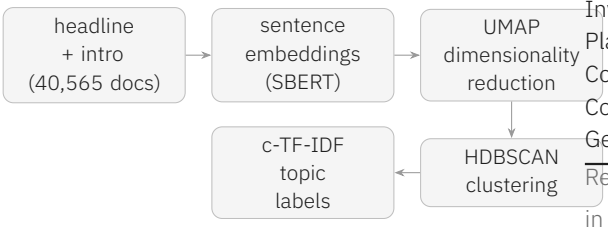
Behavioral +21.5%, enabling +13.4% in contestation corpus

BERTopic: What We Run It On

Input: 40,565 headlines + intros

From `datacentering_crawl_dcd.tsv`:

"Microsoft expands plans for La Porte, Indiana, data center campus. Company seeks to rezone..."



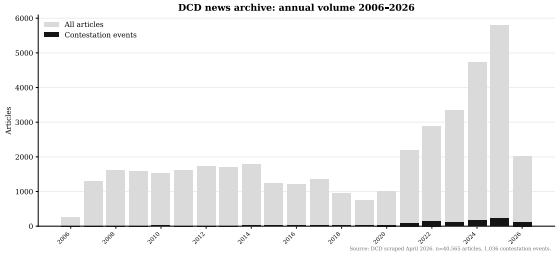
Expected topic clusters

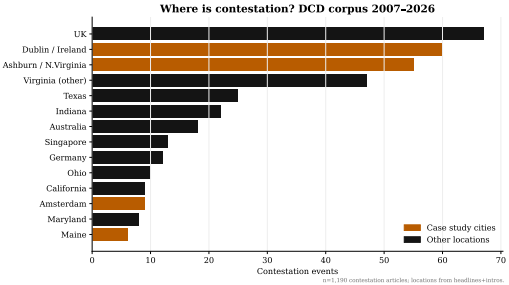
Cluster type	Example terms
Energy / grid	MW, capacity, power, grid
Investment / capex	$B, funding, lease, REIT
Planning / planning	council, rezone, permit
Contestation	moratorium, oppose, sue
Cooling / efficiency	PUE, liquid, immersion
Geography	Virginia, Ireland, Singapore

Results running — will appear in updated slides.

`data/processed/wp4_dcd/bertopic_topics.csv`

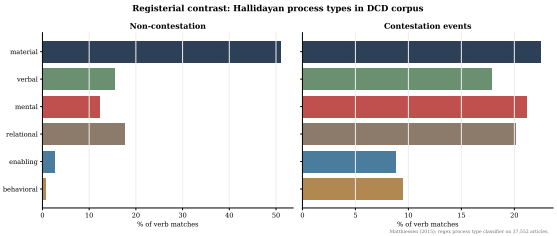
The Arena in Data





37,552 articles | 1,073 contestation events | peak 2025. **Headlines + intro text only** — full articles via screenshot/OCR pipeline. More coming.

The Register Shift



Matthiessen (2015) registerial cartography — verb process types

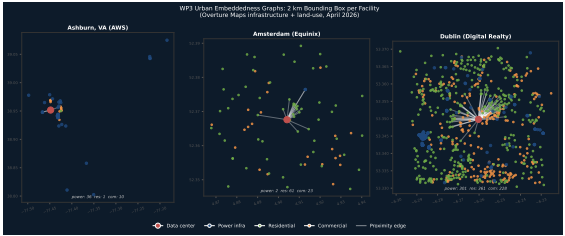
Type	Contest	Non
Behavioral	23.1%	1.6%
Enabling	16.5%	3.1%
Material	33.7%	66.8%

Doing → governing + contesting

The governance gap visible in the verb register.

Headlines + intros only.

City2Graph: Urban Embeddedness



From corpus cities → urban graphs

GLiNER NER extracts city names
→ city2graph pulls Overture Maps
→ graph: DC node ↔ power, residential, commercial

	Res. zones	Power
Ashburn	1	36
Amsterdam	61	2
Dublin	361	6

Spatial embeddedness determines governance counterparty availability.

Three Research Agendas

Agenda	Central question	Cases
Regional Innovation Systems	How do hyperscale clusters reshape local productivity, labour, energy, property?	Loudoun (US), Luleå (SE), Dublin (IE), Hyderabad (IN)
Sectoral Transformation	How does hyperscaler vertical integration reshape sectoral systems and startup access?	AWS, Google, Meta, Microsoft platform dependency
Policy Sovereignty & Energy	What policy mix sustains sovereign compute without fossil lock-in or geopolitical dependency?	UK AI Growth Zones, EU AI Factories, France, Saudi HUMAIN, UAE Stargate

Each agenda operationalized via the CITC cycle (Berente et al. 2019): sampling → synchronic → diachronic → lexical framing.

Three Contributions

Cartalog of Actors

Who is in the datacentering arena?

40,565 DCD articles → NER → actor typology: hyperscalers, planners, communities, courts, utilities

Geocoded, dated, open (CC BY 4.0)

Cartalog of Events

What happens in the arena?

1,036 contestation events → event type, location, date, actors involved

Register shift: *doing* → *enabling + contesting*

Peak: 2025 (214 events)

Methodological Toolkit

How to study the arena?

Arenas of Transformation (after Jørgensen 2012)

+ Situational maps (Clarke)
+ CITC (Berente et al. 2019)
+ Registerial cartography (Matthiessen 2015)

Replicable. Open pipelines.

Building the Cartography Together

Why open is theoretical necessity

CAS-like phenomena are too distributed, too opaque for any single team.

What we share

- Facility index (CC BY 4.0)
- NLP / CV / network pipelines (MIT)
- DCD contestation corpus
- Register + NER analysis scripts
- `github.com/babajideowoyele/datacentering-cartography`
- **Cartalog UI** — live actor network, topic map, spatial map

Concrete asks

- Local planning records + disputes
- Non-English corpus (NL, DE, FR, IE)
- Company registry / REIT data access
- Replication in other jurisdictions
- Methodological critique

*"What would innovation studies look like if we treated data infrastructure as **foundational** rather than peripheral?"*

Q: Isn’t this STS dressed up as innovation studies?

Live: “Fair point — I adapt Jørgensen’s *arenas of development* (RP 2012) into *arenas of transformation* because the question isn’t how DCs get built but what they do to the systems around them: energy grids, planning regimes, labour markets. The *transformation* framing is also consistent with transitions vocabulary your audience knows.”

The methodological spine — CITC (Berente, Seidel & Safadi 2019; Miranda et al. 2022) — comes from *MIS Quarterly* and *ISR*, not from STS. The agenda is testable, not hermeneutic.

Q: Where's the empirical contribution?

Live: "The ECR paper is the agenda. The corpus analysis is the first empirical instantiation."

From 37,552 DCD articles:

Analysis	Status	Finding
EDA geography	Complete	US/Europe split; peak 2025
Register analysis	Complete	Behavioral +21.5%, enabling +13.4%
BERTopic topics	Running	Full corpus topic clusters
city2graph	Seed run	3 cities; scales via GLiNER NER

Analysis on headlines + intros. Full article text: screenshot/OCR pipeline coming.

Q: Why Loudoun, Luleå, Dublin, Hyderabad?

Variation on three dimensions:

Case	Regulatory	Energy	Trajectory
Loudoun	Common law / NIMBY	Gas + solar	Mature US cluster
Luleå	Swedish planning	98% hydro	Renewable-led
Dublin	Irish planning	Mixed	FDI platform
Hyderabad	Indian regulatory	Expanding	Emerging hub

Q: What are the register verb families?

Following Matthiessen (2015) Hallidayan process types — applied to DCD headlines + intros:

Process type	Field of activity	Top verbs in corpus
Material	Doing / building	secures, builds, launches, withdraws, rejects
Verbal	Reporting	says, announces, claims, warns, demands
Mental	Exploring / sharing	opposes, concerns, challenges, supports
Relational	Expounding	is, becomes, remains, includes
Enabling	Enabling	bans, permits, rezones, mandates, regulates
Behavioral	Contesting	protests, campaigns, sues, appeals, petitions

Contestation = shift from material/relational to behavioral/enabling.

Q: Why CITC and not CSS more broadly?

CITC's lexical-framing requirement is what makes the output theoretical, not just descriptive.

That's the discipline that protects against the most common reviewer objection:

> *"This is interesting, but what's the theoretical contribution?"*

CITC explicitly distinguishes patterns from contributions. The test for RP.